LABSHEET-10

Q1. Display the available block devices and identify a dedicated test partition that can be used for file-system practice.

lsblk

Q2. Display the current file-system type, UUID and mount-point information for the selected test partition.

sudo blkid /dev/sdb1

Q3. Create an ext4 file system on a dedicated test partition using mkfs and verify the file-system creation.

sudo mkfs.ext4 /dev/sdb1

Q4. Display detailed information about the newly created file system and identify its UUID.

sudo tune2fs -l /dev/sdb1

or

sudo blkid /dev/sdb1

Q5. Create a directory named /mnt/labfs to use as the mount point for the test file system.

sudo mkdir -p /mnt/labfs

Q6. Mount the newly created file system manually on /mnt/labfs and verify that it is mounted correctly.

sudo mount /dev/sdb1 /mnt/labfs

findmnt /mnt/labfs

Q7. Display the mounted file system using df and identify its total, used and available space.

df -h /mnt/labfs

Q8. Create a test file inside /mnt/labfs and verify that the file is stored on the mounted file system.

echo 'Hello File System' | sudo tee /mnt/labfs/testfile.txt

ls -l /mnt/labfs

Q9. Unmount the test file system using umount and verify that /mnt/labfs is no longer an active mount point.

sudo umount /mnt/labfs

findmnt /mnt/labfs

Q10. Display the current contents of /etc/fstab and identify the fields used to define a persistent file-system mount.

cat /etc/fstab

Q11. Add an entry for the dedicated test file system to /etc/fstab using its UUID and an appropriate mount point.

UUID=$(sudo blkid -s UUID -o value /dev/sdb1)

echo "UUID=$UUID /mnt/labfs ext4 defaults 0 2" | sudo tee -a /etc/fstab

Q12. Verify the syntax and test the /etc/fstab entry without rebooting by using the appropriate mount command.

sudo mount -a

findmnt /mnt/labfs

Q13. Reboot or remount the test system as appropriate and verify that the file system is automatically mounted according to /etc/fstab.

```
sudo mount -a
df -h /mnt/labfs
```

Q14. Use findmnt and df to verify the device, mount point, file-system type and storage usage after the automatic mount.

```
findmnt /mnt/labfs
df -hT /mnt/labfs
```

Q15. Display the file-system UUID and compare it with the UUID used in the /etc/fstab entry.

```
sudo blkid /dev/sdb1
grep '/mnt/labfs' /etc/fstab
```

Q16. Run an appropriate file-system checking command on the dedicated test file system after it has been safely unmounted.

```
sudo umount /mnt/labfs
sudo fsck.ext4 /dev/sdb1
```

Q17. Demonstrate the difference between checking a mounted file system and checking an unmounted file system, using a dedicated test partition and appropriate precautions.

```
# Running fsck on a mounted filesystem risks data corruption.
# Safe execution (unmounted):
sudo umount /dev/sdb1 2>/dev/null
sudo fsck -f /dev/sdb1
```

Q18. Correct an intentionally incorrect /etc/fstab entry for the dedicated test file system and verify that the corrected entry mounts successfully.

```
# Edit /etc/fstab to fix syntax/UUID errors:
sudo nano /etc/fstab


# Test mount:
sudo mount -a
```

Q19. Unmount the test file system, remove the lab-specific /etc/fstab entry and verify that the system no longer attempts to mount it automatically.

```
sudo umount /mnt/labfs
sudo sed -i '\|/mnt/labfs|d' /etc/fstab
findmnt /mnt/labfs
```

Q20. Perform a complete file-system workflow: create a file system, mount it, verify it, configure /etc/fstab using UUID, check the file system safely, unmount it and clean up all lab changes.

```
sudo mkfs.ext4 -F /dev/sdb1
sudo mkdir -p /mnt/labfs
sudo mount /dev/sdb1 /mnt/labfs
df -h /mnt/labfs
```

```
sudo umount /mnt/labfs
sudo fsck /dev/sdb1
sudo rmdir /mnt/labfs
```